

Ex 4:

Step 1: Setup Prerequisites

Description:

- Git is installed and configured
- P4Merge tool is available for conflict resolution visualization
- Local Git repository is connected to a GitLab remote repository

Step 2: Verify Clean Master State

git checkout master

git status

- Output should indicate "nothing to commit, working tree clean"

Step 3: Create a Branch and Add File

git branch GitWork

git checkout GitWork

echo "<message>Hello from branch</message>" > hello.xml

git add hello.xml

Step 4: Commit Branch Changes

git commit -m "Added hello.xml in GitWork branch"

Step 5: Switch to Master and Make Conflicting Change

git checkout master

echo "<message>Hello from master</message>" > hello.xml

git add hello.xml

git commit -m "Added hello.xml in master with different content"

Step 6: View Logs

git log --oneline --graph --decorate --all

Step 7: Compare Differences

In CLI:

```
git diff master GitWork
```

In P4Merge:

```
git mergetool
```

Step 8: Merge and Trigger Conflict

```
git merge GitWork
```

- Git will report a **merge conflict** in hello.xml

Step 9: Resolve Conflict with 3-Way Merge Tool

```
git mergetool
```

- In P4Merge, choose the appropriate changes from **master**, **branch**, or **both**, then save.

Step 10: Commit Resolved Merge

```
git add hello.xml
```

```
git commit -m "Resolved merge conflict in hello.xml"
```

Step 11: Ignore Backup Files

```
echo "*.orig" >> .gitignore
```

```
git add .gitignore
```

```
git commit -m "Updated .gitignore to exclude backup files"
```

Step 12: Delete Branch After Merge

```
git branch -d GitWork
```

Step 13: View Final Log

```
git log --oneline --graph --decorate
```

```

-<message>Hello from MAIN branch</message>
+<message>Hello from GitWork branch</message>

Thanisha G@LAPTOP-SACJTKTA MINGW64 ~/GitDemo (main)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

Thanisha G@LAPTOP-SACJTKTA MINGW64 ~/GitDemo (main|MERGING)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both added:      hello.xml

no changes added to commit (use "git add" and/or "git commit -a")

Thanisha G@LAPTOP-SACJTKTA MINGW64 ~/GitDemo (main|MERGING)
$ "C:/Program Files/Notepad++/notepad++.exe" hello.xml

Thanisha G@LAPTOP-SACJTKTA MINGW64 ~/GitDemo (main|MERGING)
$ git add hello.xml

Thanisha G@LAPTOP-SACJTKTA MINGW64 ~/GitDemo (main|MERGING)
$ git commit -m "Resolved merge conflict"
[main d846655] Resolved merge conflict

Thanisha G@LAPTOP-SACJTKTA MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 5 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

Thanisha G@LAPTOP-SACJTKTA MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate
*   d846655 (HEAD -> main) Resolved merge conflict
| \
|  * e70e62d (Gitwork) Added hello.xml in GitWork branch
| * | 60a21a2 Added hello.xml in main branch
|/
* f7bafec Added branch file
* 956cc21 Added branch file
* d0af2cc (origin/main) Added .gitignore
* 35328c4 Initial commit

Thanisha G@LAPTOP-SACJTKTA MINGW64 ~/GitDemo (main)
$ git branch -d GitWork
Deleted branch GitWork (was e70e62d).

Thanisha G@LAPTOP-SACJTKTA MINGW64 ~/GitDemo (main)
$

```